

5.48 $m = 100 \text{ kg/hr}$

✓ 25% accl

✓ 75% water

pure solvent

in BL

6% accl 2% water

0.2% solvent

67.4% accl

27% in BL

92% water

x

$$25 \text{ kg} - \frac{0.06 \cdot m_4}{0.06} = 0.06 m_4$$

$$25 \text{ kg} - m_4 = m_3$$

$$416.6 \text{ kg} - m_4 = m_3$$

$$75 \text{ kg} = 0.02 \cdot (416.6 - m_4) +$$

$$0.42 \cdot m_4$$

$$75 = 8.332 - (0.02 \cdot m_4) + 0.42 m_4$$

$$-8.332$$

$$66.668 = -0.02 + 0.42$$

$$\frac{66.668}{0.40} = -0.02 + 0.42 = \frac{66.668}{0.40} = 166.67$$

$$m_4 = 166.67$$

$$25 \text{ kg accl} \quad \checkmark 25 \text{ kg in} = 0.06 \cdot m_3 + 0.06 \cdot m_4$$

$$75 \text{ kg water} \quad \checkmark 75 \text{ kg in} = 0.02 \cdot m_3 + 0.42 \cdot m_4$$

$$m_2 = (0.42 \cdot m_3) + (0.02 \cdot m_4)$$

$$\frac{66.668}{m_4} = -0.02 + 0.42 = \frac{66.668}{m_4} = 0.40 = \frac{66.668}{0.40} = 166.67$$

$$m_4 = 166.67$$

$$25 = 0.06 \cdot m_3 + 4.44$$

$$-4.44 \quad \frac{0.06}{0.06} = m_3 \cdot 0.06$$

$$20.56 = m_3 \cdot 0.06$$

$$\frac{20.56}{0.06} = m_3 \cdot 0.06$$

$$342.6 = m_3$$

$$100 \text{ kg} + m_2 = 342.6 + 166.67$$

$$508.67 \text{ kg}$$

$$\frac{316.67}{100} = 3.1667 \text{ kg solvent per kg feed}$$

$$3.1667$$

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missing $f_{ad} = 1$.
 plots ~ 2
 18/05

#2/5.41

from fit y vs x

$$slope = 0.2469$$

$$intercept = 0.1241$$

$$q_{\infty} = \frac{1}{0.2469} = 3.37 \text{ mmol/g}$$

$$k_L = \frac{0.2469}{0.1241} = 2.39 \text{ L/mol}$$

$$q_{MLP} = \frac{(2.39)_{LP}}{1 + (2.39)_{LP}}$$

$$R^2 = 0.89$$

with:

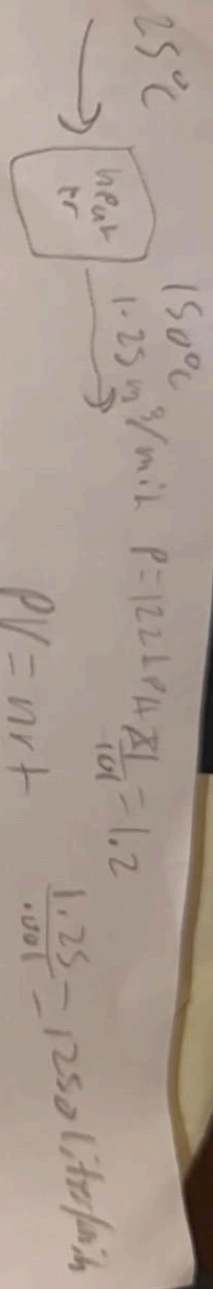
$$n = \frac{1}{3} = \frac{1}{0.2469} = 4.12$$

$$q_{PLCP} = 2.14 \text{ g/g}$$

$$R^2 = 0.99$$

Freundlich is the better model

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$$\Delta H = 3640 \text{ J/mol}$$

$$R = 1.08206 \text{ atm liter} / \text{mol K}$$

$$PV = nRT$$

$$n_2 = n_1$$

$$n_2 = 43.9 \text{ mol/min} \quad 150^\circ\text{C} = 423 \text{ K}$$

$$n_2 = 43.9 \text{ mol/min} \cdot \frac{1 \text{ min}}{60 \text{ sec}} = 0.72 \text{ mol/sec} \quad \checkmark$$

$$E = n \cdot \Delta H + 0.72 \cdot 3640 = 2632.43 \text{ J/sec}$$

$$\frac{2632 \text{ kJ/sec} = \text{kJ}}{1000} \quad \checkmark$$

if you heat the gas up its going to increase KE of the gas coming out, ΔE must be positive. ✓

2/3/19

4
a) $1 \text{ kg m/sec}^2 = \text{J}$

$$\Delta E_k = \frac{1}{2} m (V_{\text{final}} - V_{\text{initial}})$$

$$100 \cdot \frac{5}{18} = 27.77$$

$$(27.77)^2$$

$$\frac{23}{25}$$

$$\Delta E_k = \frac{1}{2} m \cdot (771.6)$$

g mol = 18

$$\frac{\frac{1}{2} \cdot 18}{1000}$$

$$\cdot (771.6) = 6.94 \text{ J} \quad \checkmark$$

b) $\Delta E = \frac{18}{1000} \cdot 9.8 \cdot 100 = 17.64 \text{ J} \quad \checkmark$

c) b-16 $1 \text{ bar } 100^\circ\text{C} = 2687 \text{ kJ/kg}$ $1 \text{ bar } 100^\circ\text{C} = 2675.84 \text{ kJ/kg}$

$$\Delta H = -11.2 \quad \Delta E = m \cdot \Delta H = -0.2016 \text{ J} \quad \checkmark$$

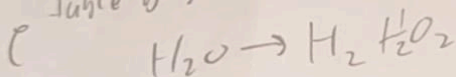
$$\Delta E = n \int_{T_1}^{T_2} c_p dt = \frac{1}{18} \cdot \int_{273}^{373} (a + bT) dT = \int_{273}^{373} (22.4 + 0.0104(T)) dT$$

$$\left(\frac{1}{18} \cdot \left(22.4T + \frac{0.0104 \cdot T^2}{2} \right) \right) \Big|_{273}^{373}$$

$$\rightarrow 2576 \text{ J} \quad \checkmark$$

$$T = ?$$

Table b-3



$$\Delta H_f = H_{f, \text{H}_2} + H_{f, \text{O}_2} - H_{f, \text{H}_2\text{O}}$$

$$\Delta H_f = 0 + 0 + 2.86 \times 10^3$$

$$\Delta E = \frac{18}{1000} \cdot 2.86 \times 10^3 = 51.48 \text{ J} \quad \times$$

$$\left(\frac{18}{1000} \cdot 3 \times 10^8 \right)^2 = 9 \times 10^{16} = \boxed{1.62 \times 10^{15} \text{ J}} \quad \checkmark$$